EC9600: Applied Algorithms

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Combinatorial Algorithms and Graph Theory

Outline

- Generating all and random instances of a combinatorial object;
- Exhaustive search through the solution space, which are represented as combinatorial structures such as permutations, combinations, set partitions, integer partitions, and trees;
- Graph theoretic models

Combinatorial Algorithms

- Combinatorial object is a discrete (non-continuous) structure built from a finite set of elements according to specific rules or patterns.
- Widely used in routing and scheduling.
- Examples:
  - **Permutations:** Distinct ordered arrangements of a set.
  - **Combinations / Subsets:** Unordered selections of elements from a larger set.
  - **Graphs & Trees:** Collections of nodes (vertices) connected by edges, used to model networks, relationships, or hierarchies.
  - **Partitions:** Ways to split a set into non-empty parts.
  - **Latin Squares & Sudoku Grids:** Matrices filled with symbols under strict row and column constraints.
- Key Characteristics:
  - **Discrete & Finite:** Made up of individual, distinct elements (e.g., integers, graph nodes, letters).
  - **Structured:** Arranged according to well-defined rules or constraints.

Permutations

- Domain used:
  - **Routing Problem:** A class of optimization problems focused on determining the most efficient routes for transportation or service delivery, minimizing travel time, distance, or cost.
    - **Vehicle Routing Problem (VRP):** A widely studied combinatorial optimization problem that involves designing optimal routes for a fleet of vehicles serving a set of customers with specific constraints.
    - **Traveling Salesman Problem (TSP):** A classic routing problem in which a salesperson must visit a set of cities exactly once and return to the starting point while minimizing the total travel distance or cost
- Applications:
  - Logistics and Supply Chain: Optimizing delivery routes for e-commerce and last-mile delivery.
  - Public Transportation: Designing efficient bus and train schedules.
  - Healthcare Logistics: Optimizing emergency medical services (EMS) and vaccine distribution.
  - Smart Cities: Managing traffic flow and waste collection routes.

Travelling Salesman Problem (TSP)

- We are given n vertices 1, . . . , n and all n(n − 1)/2 distances between them, as well as a budget b. We are asked to find a tour, a cycle that passes through every vertex exactly once, of total cost b or less or to report that no such tour exists.

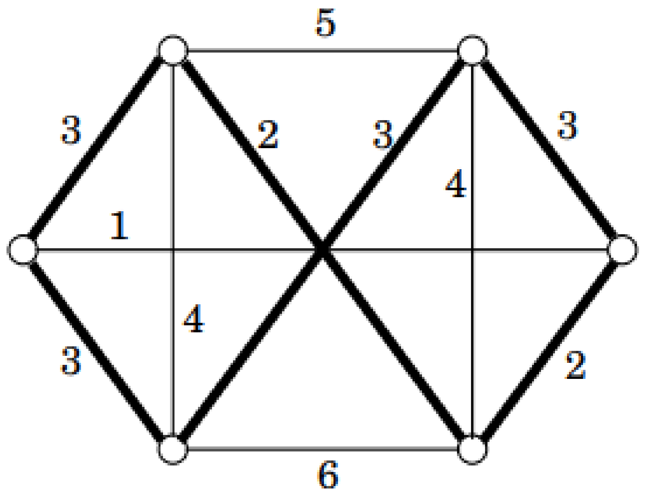

Figure 1: The optimal traveling salesman tour;

Source: https://people.eecs.berkeley.edu/~vazirani/algorithms/chap8.pdf

Combinations

- Domain used:
  - capacitated vehicle routing problem (CVRP)
  - team orienteering problem (TOP)
- Applications:
  - load distribution
  - clustering deliveries
  - reducing overall travel costs

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Heuristics

- Methods that cannot be guaranteed to produce optimal solutions, but we hope that it will produce approximately good solutions.
- Two types of heuristics:
    - Attempts to construct a good solution from scratch
    - Tries to improve an existing solution (local improvement)

Exercise - 01

The table below shows the distances, in km, between six data collection points, A, B, C, D, E, and F. Rachel must visit each collection point. She will start and finish at A and wishes to minimize the total distance travelled.

	A	B	C	D	E	F
A	-	77	34	56	67	21
B	77	-	58	58	36	74
C	34	58	-	73	70	42
D	56	58	73	-	68	38
E	67	36	70	68	-	71
F	21	74	42	38	71	-

Exercise - 01

1. Starting at A, use the **nearest neighbour** algorithm to obtain an upper bound. Make your method clear. Starting at B, a second upper bound of 293 km was found.

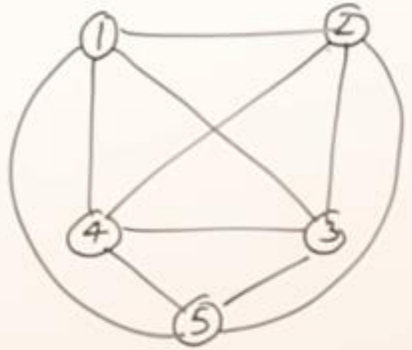
2. State the better upper bound of these two, giving a reason for your answer. By deleting A, a lower bound was found to be 245 km.

3. By deleting B, find a second lower bound. Make your method clear. State the better lower bound of these two, giving a reason for your answer.

4. Use inequalities to write down an interval that must contain the length of Rachel's optimal route

Exercise - 02

Find the minimal path from A using branch and bound technique for TSP. The algorithm is discussed with steps in class.



$$
\begin{array}{c c}
 & \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \end{array} \\
\begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{array} &
\begin{bmatrix}
\infty & 20 & 30 & 10 & 11 \\
15 & \infty & 16 & 4 & 2 \\
3 & 5 & \infty & 2 & 4 \\
19 & 6 & 18 & \infty & 3 \\
16 & 4 & 7 & 16 & \infty
\end{bmatrix}
\end{array}
$$

Reference books

1. A. Nayak and I. Stojmenović, Eds., “Handbook of Applied Algorithms,” Mar. 2007, doi: 10.1002/9780470175668.
2. A. De, M. Gorton, C. Hubbard, and P. Aditjandra, “Optimization model for sustainable food supply chains: An application to Norwegian salmon,” Transportation Research Part E: Logistics and Transportation Review, vol. 161, p. 102723, May 2022, doi: 10.1016/j.tre.2022.102723.